Aspect	Status
Security detection	❌ Broken

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15. If you want next level

I can help you build:

1. Fully correct BB84 simulation

- with noise
- with proper interception

2. Visualization

- error rate vs Eve presence

3. Research-level extension

- decoy states
- realistic channel noise

Before proceeding, answer this:

Do you want:

1. **Correct working BB84 code**
2. **Deep theory (security proof level)**
3. **Project-level simulation with graphs**

Choose one.

i didnt get anything just explain all with proper theory please

Understood. I'll rebuild this from first principles, with minimal assumptions and no code distractions.

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1. What problem BB84 solves

The **BB84 Protocol** is used to **share a secret key securely** between two parties:

- Alice → sender

- Bob → receiver
- Eve → eavesdropper

Goal:

Create a key such that:

- Alice and Bob share the same bits
- If Eve tries to listen → she is detected

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2. Classical vs Quantum Security (why BB84 exists)

Classical communication:

- Eve can copy data without changing it
- ❗ No way to detect eavesdropping

Quantum communication:

- Measuring a quantum state **changes it**
- ❗ Eve cannot observe without disturbing

👉 This is the entire foundation of BB84

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3. Quantum States used

BB84 uses **2 bases** (think of them like 2 languages)

(A) Z-basis (standard)

Bit	State
0	
1	

(B) X-basis (rotated)

Bit	State
0	

Bit	State
1	

These states are:

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

$$|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

Important property

👉 If you measure in the **wrong basis**, result becomes random

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4. Full BB84 Protocol (step-by-step)

Step 1: Alice prepares data

She randomly chooses:

- Bit (0 or 1)
- Basis (Z or X)

Example:

Index	Bit	Basis
1	0	Z
2	1	X
3	0	X

Step 2: Alice encodes qubits

She converts each (bit, basis) into a quantum state:

Bit	Basis	State
0	Z	

Bit	Basis	State
1	Z	
0	X	
1	X	

Step 3: Alice sends qubits to Bob

👉 These are **quantum particles (photons in real systems)**

Step 4: Bob measures

Bob randomly chooses basis:

- Z or X

Two cases:

✔️ Case 1: Bases match

- Alice = Z, Bob = Z
- OR Alice = X, Bob = X

👉 Bob gets correct bit

❌ Case 2: Bases differ

- Alice = Z, Bob = X
- OR Alice = X, Bob = Z

👉 Bob gets random result

---

5. Basis comparison (public step)

After transmission:

- Alice and Bob share their bases (NOT bits)
- They keep only positions where bases match

👉 This is called **sifting**

Result:

They now have a **shared key**

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6. Where Eve comes in

Eve tries to intercept:

1. She measures qubit
2. She sends new qubit to Bob

Problem for Eve

👉 She does NOT know the correct basis

So she guesses:

- 50% chance correct
- 50% chance wrong

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7. What happens when Eve is wrong

Let's trace:

Example

Alice sends:

- Bit = 0
- Basis = X → state = $|+\rangle$

Eve measures in Z basis (wrong)

👉 Result becomes random:

- Could be 0 or 1

Eve sends new qubit

Now the original information is lost

Bob measures correctly

👉 But state is already corrupted

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8. Error Rate (very important)

Because of Eve:

- Wrong basis → introduces errors

Final result:

$$\text{Error Rate} \approx 25\%$$

Why 25%?

- 50% chance Eve chooses wrong basis
- In those cases → 50% error

👉 So:

$$0.5 \times 0.5 = 0.25$$

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9. How Alice & Bob detect Eve

They:

1. Take a small subset of shared key
2. Compare publicly

If error rate:

Error Rate	Meaning
~0%	No Eve

Error Rate	Meaning
~25%	Eve present

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10. Why this is secure (core theory)

Security is based on:

👉 Heisenberg Uncertainty Principle

Meaning:

- You cannot measure a quantum system without disturbing it
- Especially when basis is unknown

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11. Connecting to your code (simple mapping)

Alice

```python
alice_bits → bit
alice_bases → basis
```

Encoding

```python
X gate → bit = 1
H gate → change basis  Z
```

Bob

```python
H before measurement → measure in X basis
No H → measure in Z basis
```

12. Why your output showed 0 error

👉 Because:

- Eve was not properly modifying qubits
- Bob effectively received original states

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13. One-line intuition

👉 BB84 =
"Wrong measurement destroys information, and that destruction reveals eavesdropping"

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14. Short viva summary

- Uses 2 bases (Z and X)
- Random encoding and measurement
- Matching bases → correct bits
- Mismatched bases → random
- Eavesdropping introduces errors
- Error rate reveals attacker

If you still feel confused

Tell me specifically:

- "I don't understand basis"
- "I don't understand measurement"
- "I don't understand Eve effect"

I will break that single concept even more simply.